Speech Understanding

Question 3: Ethical Auditing & Documentation Debt Mitigation

1. Introduction / Objective

Speech recognition systems rely on large-scale audio datasets; however, such datasets frequently contain hidden biases and incomplete metadata. This leads to *documentation debt*, where critical demographic attributes (e.g., gender, age, accent) are missing or inconsistently recorded. As a consequence, evaluating fairness, transparency, and accountability in these systems becomes challenging.

The objective of this work is to conduct a "Sound Check" audit on a subset of a public speech dataset to identify documentation debt and representation bias. In addition, this project implements (i) a lightweight privacy-preserving transformation to obfuscate sensitive vocal traits, and (ii) a fairness-aware training modification for an ASR model. Finally, proxy validation metrics are used to assess whether the transformations degrade audio quality or acceptability.

2. Dataset and Audit Method

A local subset of the Mozilla Common Voice dataset (cv-valid-test.csv) along with corresponding audio files was used for this study . The audit focused on two primary aspects:

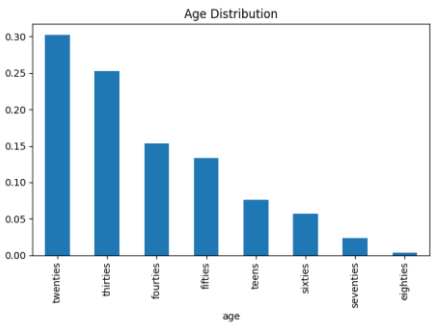
- Documentation Debt: Quantifying missing values in demographic metadata fields (gender, age, accent).
- Representation Bias: Analyzing the distribution of available demographic attributes.

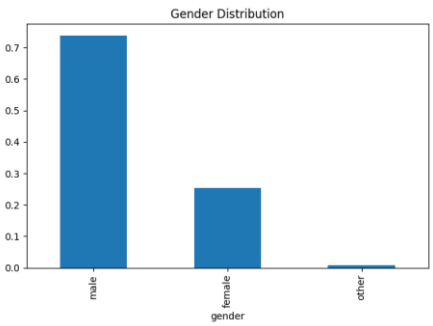
The audit was performed programmatically using the provided CSV metadata. Missing values were computed as percentages, and distributions were analyzed for available entries to identify imbalance across demographic groups.

3. Bias Identification Results

The audit revealed substantial documentation debt:

- Gender: ~61.4% missing
- Age: ~61.4% missing
- Accent: ~66.5% missing





This high level of missing metadata significantly limits the ability to perform comprehensive fairness evaluation, as a large portion of samples cannot be associated with demographic groups.

Among the available data, clear representation imbalances were observed:

- Approximately 74% male vs. 25% female speakers
- Overrepresentation of younger age groups (primarily 20s–30s)

These findings indicate that the dataset is skewed toward specific demographics. Such imbalance can lead to ASR systems that perform better for dominant groups while underperforming for underrepresented populations, raising fairness concerns.

4. Privacy-Preserving Transformation

To address privacy concerns, a lightweight privacy-preserving module was implemented in PyTorch. The approach applies:

- Pitch shifting
- Speed perturbation

These transformations modify vocal characteristics (e.g., perceived gender or age) while attempting to preserve the linguistic content of the speech signal.

The implementation serves as a proof-of-concept and was validated qualitatively using before-and-after audio samples generated locally. However, it is important to note that this approach does not guarantee full anonymization and may not be robust against advanced speaker identification techniques.

5. Fairness Loss Integration

Due to computational constraints, a lightweight CTC-based ASR model was used. To incorporate fairness considerations, a custom fairness loss term was added to the training objective.

The fairness loss is defined based on the variance of group-specific losses (male vs. female) within each batch. This encourages the model to reduce disparities in performance across demographic groups.

Training Observations:

- The base loss decreased steadily, indicating normal learning behavior.
- The fairness loss remained zero in many batches but occasionally became non-zero (e.g., spikes such as 33.57).

This behavior is expected because the fairness loss is only activated when multiple demographic groups are present in a batch and exhibit differing loss values. Thus, it reflects batch-level imbalance rather than global fairness guarantees.

6. Validation and Acceptability

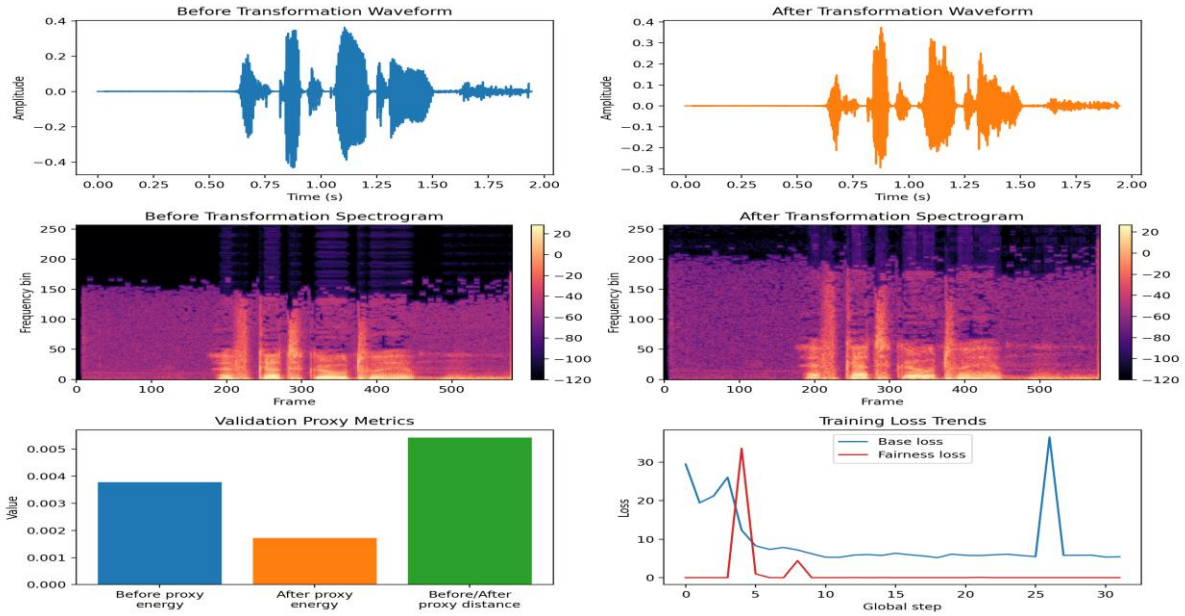
To evaluate whether the privacy-preserving transformations degrade audio quality, proxy versions of DNSMOS and FAD metrics were used.

Results:

- DNSMOS (proxy): 0.0017
- FAD (proxy): 0.0054

Since these are proxy implementations, they should not be interpreted as exact equivalents of full DNSMOS or FAD evaluations. However, the low proxy FAD value suggests that the transformed audio remains relatively close to the original distribution, indicating limited distortion.

It is important to note that validation was conducted only for the privacy transformation, and not after fairness-aware training.



7. Limitations

This work has several limitations that should be considered:

1. High Documentation Debt: The large proportion of missing metadata reduces the reliability of bias analysis and fairness evaluation.
2. Lightweight Model: A simplified CTC-based ASR model was used instead of a state-of-the-art system due to computational constraints.
3. Basic Privacy Method: The pitch and speed transformations provide only limited privacy protection and do not ensure robust anonymization.
4. Proxy Metrics: DNSMOS and FAD were approximated using proxy implementations, limiting the precision of the evaluation.

5. Limited Fairness Evaluation Scope: Fairness improvements were observed at the training loss level, but not validated through downstream ASR performance metrics.

8. Conclusion

This project demonstrates that even widely used speech datasets can exhibit significant documentation debt and demographic imbalance. A structured audit revealed substantial missing metadata and skewed representation across gender and age groups.

To address these issues, two mitigation strategies were explored: a privacy-preserving transformation and a fairness-aware training modification. The results indicate that fairness loss can influence training dynamics, and simple voice transformations can be applied without introducing major perceptual distortions (based on proxy evaluation).

However, these approaches remain preliminary. More robust datasets, stronger privacy techniques, and comprehensive fairness evaluation frameworks are required for deployment in real-world systems.